Name_____ Date_____ Period_____

LESSON 4-1 · SESSION 1 · GUIDED NOTES

Percent Means Per Hundred

Learning target – I can write a ratio as a percent by finding the equivalent ratio out of 100. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
percent *por ciento*	Out of ____, or per hundred.	40% = 40 out of 100
decimal grid *cuadrícula decimal*	A grid of 100 squares. Each square is ___.	62 shaded = 62%
equivalent ratio *razón equivalente*	Shows the _____ comparison with different numbers.	2 : 5 = 40 : 100

KEY IDEA

complete each sentence

- Percent means per _____. The whole is always _____.
- Multiply the part and the whole by the _____ number to make the whole 100.

WORKED EXAMPLE

fill in the blanks as we solve

3 out of every 4 students bring lunch from home. What percent bring lunch?



Step 1 What times 4 makes 100? ___.

Step 2 Multiply both: 3 × 25 = ___ and 4 × 25 = 100.

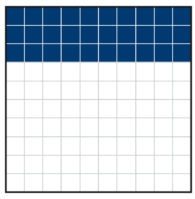
Step 3 75 out of 100 = 75%.

Answer **____ of the students bring lunch.**

YOU TRY

use the example as a model

1. The grid has 100 equal squares.



a. What percent is shaded? _____ %

b. What percent is not shaded? _____ %

2. Write each ratio as a percent.

a. 9 out of 10 = ____ %

b. 7 out of 20 = ____ %

SUM IT UP

in your own words

To write a ratio as a percent, I ___ so that the whole is 100.

Name_____ Date_____ Period_____

LESSON 4-1 · SESSION 2 · GUIDED NOTES

Percents Greater Than 100%

Learning target – I can use percents greater than 100% to describe an amount larger than the whole. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
the whole *el entero*	The amount that is _____.	last year's total = 100%
times as much *veces tanto como*	How many _____ you have.	3 times as much = 300%
greater than 100% *mayor que 100%*	_____ than the whole.	150% = 1.5 wholes

KEY IDEA

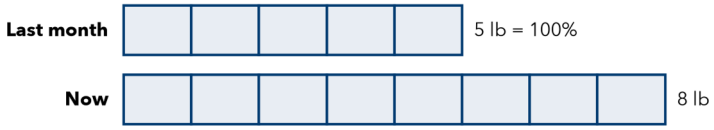
complete each sentence

- 100% is one _____. Over 100% is more than one whole.
- Find the _____ first. The whole is 100%.

WORKED EXAMPLE

fill in the blanks as we solve

A puppy weighed 5 pounds last month. Now it weighs 8 pounds. 8 pounds is what percent of 5 pounds?



Step 1 The whole: 5 pounds = _____.

Step 2 1 pound = 100% ÷ 5 = _____.

Step 3 8 pounds = 8 × 20% = 160%.

Answer **8 pounds is _____ of 5 pounds.**

YOU TRY

use the example as a model

1. Write each amount as a percent of the whole.

a. 2 times as much = _____ %

b. 1.6 times as much = _____ %

2. A class read 30 books in May. They read 45 books in June. May is the whole. June is what percent of May?

Show your work.

Answer_____ %

SUM IT UP

in your own words

A percent over 100% means the amount is ___ than one whole.

Name_____ Date_____ Period_____

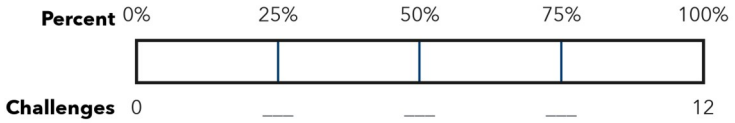
LESSON 4-1 · GUIDED NOTES · MODEL, REASON, EXPLAIN

Model It. Explain It.

SESSION 1 · PERCENT MEANS PER HUNDRED

draw it · solve it · explain it

A team finished 9 of 12 challenges at field day. Use the bar to show what percent of the challenges they finished.



a. Write the missing numbers under the bar. Shade 9 challenges.

b. What percent of the challenges did the team finish?

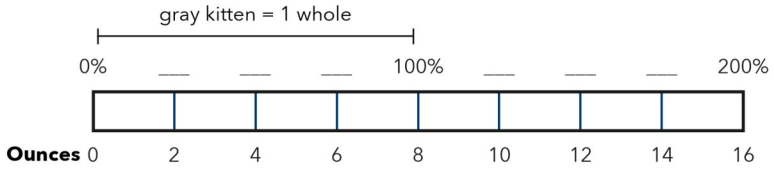
c. Explain how you know.

Challenge *Sam says 9 of 12 challenges is 9%. Use the bar to show why that is too small.*

SESSION 2 · PERCENTS GREATER THAN 100%

draw it · solve it · explain it

A gray kitten weighs 8 ounces. Its brother weighs 14 ounces. Use the bar to show the brother's weight as a percent of the gray kitten's weight.



a. Fill in the percents. Shade the bar up to 14 ounces.

b. The brother weighs what percent of the gray kitten's weight?

c. Explain how you know.

Challenge *Ali says the brother weighs 14% of the gray kitten's weight. Use the bar to show why that is too small.*

Check: Does my model match my numbers? Did I label units? Does my answer make sense?